

JavaScript

2022

Include in html

Inline script: inside `<script>` tags, or

Import script file before the end of the `<body>` (so the page is loaded, before we load the script):

```
<script src="script.js"></script>
```

Commands in html

`alert`, `prompt`, `confirm`

Notation

Camel case: `firstName`

Variable types

- Number
- String
- Boolean
- Undefined
- Null
- Symbol (cannot be changed)
- Big Int: for large integers

Variable Declaration

`let`: mutable variables

`const` : primitive, constant variables

`var`: deprecated

`2**3` means 2^3

Strings

``I am ${firstName}, a ${year - birthYear} year old.``

Curly braces define placeholders. Those quotes `` are called *backticks*.

We can print a multiline string like this:

```
console.log(`String in  
    multiple  
    lines`);
```

Conversion

Javascript uses *coercion* and automatically converts from one type to another.

```
Number("2001")
```

returns the number 2001.

NaN stands for "not a number" (invalid number)

```
String(23)
```

returns the string "23".

falsy values

These are values which return false when they are converted into boolean:

0

''

undefined

null

NaN

For example,

```
Boolean(0) = false
```

Strict Mode

We include

```
'use strict';
```

at the beginning of the script.

Functions

We can declare a function in three ways:

function declaration

```
function fName(args) {
```

```
}
```

function expression (or anonymous function)

```
const result = function(args) {
```

```
}
```

Arrow function (like Lambda expressions in C#)

```
const result = par => {do something with par};
```

```
const result = (par1, par2) => {do something with par1, par2};
```

Arrays

```
const friends = ['Nassos', 'Dimitris', 'Maria'];
```

```
const friends = new Array('Nassos', 'Dimitris', 'Maria');
```

An array may consist of values of different type, and/or functions.

Methods

- **push**: inserts a new element at the end of the array
- **unshift**: inserts a new element at the beginning of the array
- **pop**: removes the element at the end of the array and returns it.
- **shift**: removes the element at the beginning of the array.
- **indexOf**: returns the index of the element
- **includes**: returns true if the element is included in the array; otherwise, false.

Objects

Like the dictionary of Python, or the JSON type:

```
const Person = {  
    firstName: 'Nassos',  
    lastName: 'Kranidiotis',  
    age: 2021-1970,  
    job: 'IT',  
    friends: ['Nancy', 'Athina', 'Kostas']  
};
```

The elements in **red** are the properties of the object.

We call an object's property like this:

```
Person.age
```

or

```
Person['age']
```

In this way, a new property of an array can be added.

Method of an object

Methods can be declared in two ways:

1. Inside the object's block
2. Outside the object's block like the properties.

Keyword this

Inside a method, the keyword 'this' is used for calling the current object.

Print the object's properties and methods

```
console.table(object);
```

DOM

Document Object Model

All tags of an html page define a tree. DOM is not a part of Javascript. It is a part of the browser's WEB API.

Selectors

```
document.querySelector('.myclass').textContent
document.querySelector('.myclass').value
document.querySelector('.myclass').src
document.querySelector('#myid').textContent
document.querySelector('body').style.backgroundColor = ''
document.querySelectorAll...

.classList.add/remove/contains/toggle
```

Event Listeners

```
document.querySelector('.check').addEventListener('click', function(){
    [implementation]
});
document.querySelector('.check').addEventListener('click', callingFunction);
```

CallingFunction is not called immediately, so it is written without parentheses.

Keyboard Events

Keyboard events are global events.

```
document.addEventListener('keydown', function(e){
    e.key
})
```

Scope

All variables declared inside a block { } are **local variables**. Local variables are declared with the use of let or const. Local variables cannot be declared with var.

JSON

```
var jsonPerson = '{"first_name":"billy", "age":23}';
var personObject = JSON.parse(jsonPerson); //parse json string into JS object

var obj = ...;
var json = JSON.stringify(obj);
var obj2 = JSON.parse(json);
```